# Malicious PDF Detection Report

File: _upload_CS_GATE2027_Syllabus.pdf | Generated: 2026-10-05 10:59:00

## 1. File Overview

| | |
|---|---|
| File Name | _upload_CS_GATE2027_Syllabus.pdf |
| File Size | 150,835 bytes |
| SHA-256 | 81b23f40417bc1c2069782db7ad5ea01... |
| PDF Version | PDF-1.7 |
| Page Count | 2 |
| Analysis Time | 2026-10-05 10:59:00 |

## 2. File Validation

**VALID**

## 3. Risk Assessment

**32 / 100**

**SUSPICIOUS**

Confidence: Medium

## 4. Machine Learning Prediction

| | |
|---|---|
| Model | XGBoost |
| Malicious Prob. | 40.7% |
| ML Score (/ 70) | 28.5 |
| Rule Score (/ 30) | 4.0 |
| Final Risk Score | 32 / 100 |

## 5. Rule-Based Analysis

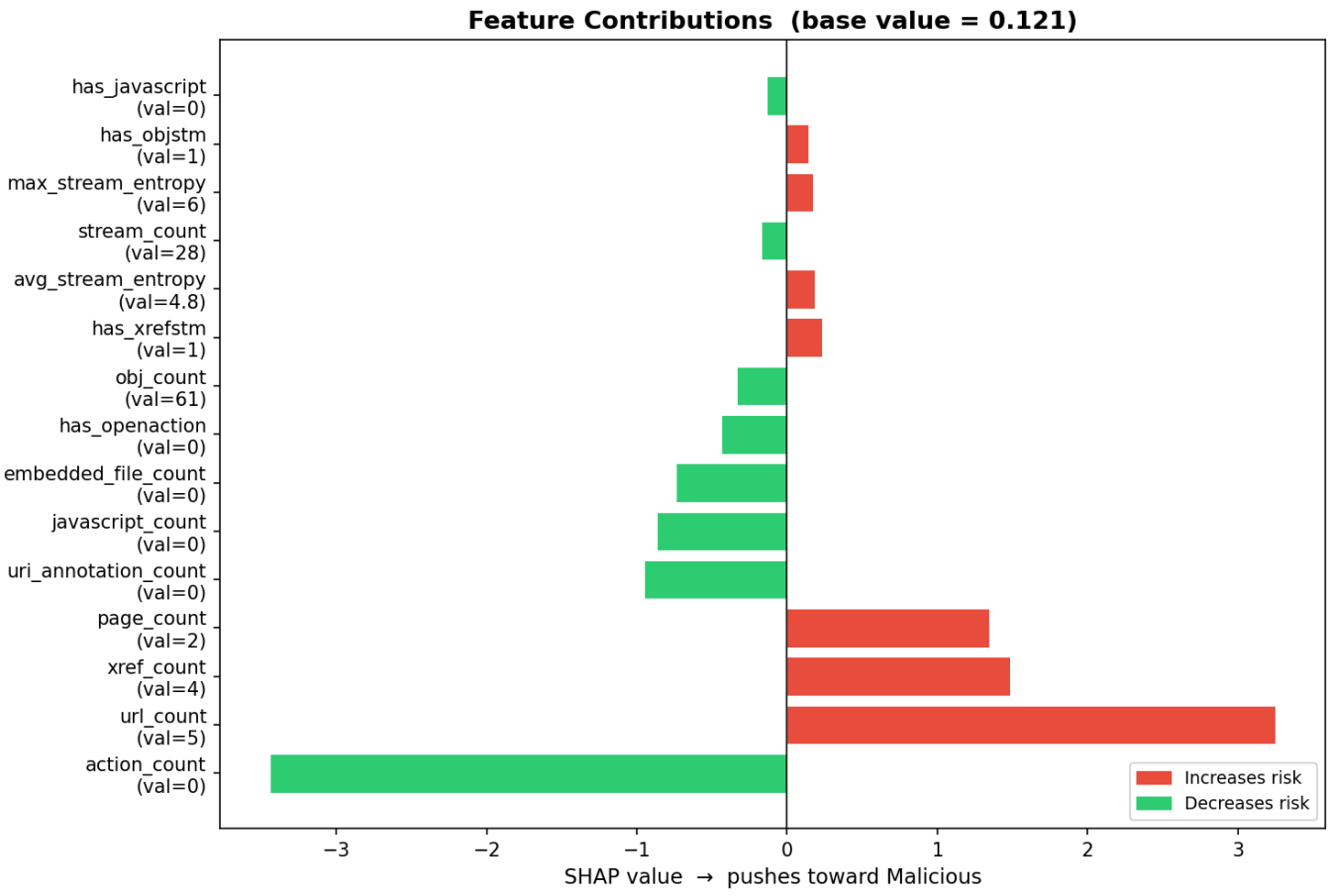
| Rule Description | Weight | Value |
|---|---|---|
| Object streams (potential obfuscation) | 0.6 | 1 |
| Multiple URLs extracted | 0.4 | 5 |
| XRef stream present | 0.3 | 1 |

## 6. Static Analysis Feature Highlights

| | |
|---|---|
| Has Javascript | 0 |
| Javascript Count | 0 |

| | |
|---|---|
| Has Openaction | 0 |
| Has Aa | 0 |
| Has Launch | 0 |
| Action Count | 0 |
| Embedded File Count | 0 |
| Has Richmedia | 0 |
| Url Count | 5 |
| Has Suspicious Url | 0 |
| Annotation Count | 0 |
| Uri Annotation Count | 0 |
| Has Acroform | 0 |
| Has Xfa | 0 |
| Is Encrypted | 0 |
| Has Objstm | 1 |
| Has Xrefstm | 1 |
| Max Stream Entropy | 5.9541 |
| Avg Stream Entropy | 4.7734 |
| High Entropy Stream Count | 0.0 |

## 7. SHAP Feature Contribution



## SHAP: Top Feature Contributions

| Feature | SHAP Value | Actual Value |
|---|---|---|
| Action Count | -3.4410 | 0 |
| Url Count | +3.2493 | 5 |
| Xref Count | +1.4821 | 4 |
| Page Count | +1.3470 | 2 |
| Uri Annotation Count | -0.9546 | 0 |
| Javascript Count | -0.8669 | 0 |
| Embedded File Count | -0.7384 | 0 |
| Has Openaction | -0.4394 | 0 |
| Obj Count | -0.3383 | 61 |
| Has Xrefstm | +0.2314 | 1 |
| Avg Stream Entropy | +0.1846 | 4.77 |
| Stream Count | -0.1744 | 28 |
| Max Stream Entropy | +0.1741 | 5.95 |
| Has Objstm | +0.1445 | 1 |
| Has Javascript | -0.1389 | 0 |

## 8. Extracted URLs (first 20)

- http://www.w3.org/1999/02/22-rdf-syntax-ns#
- http://ns.adobe.com/pdf/1.3/
- http://purl.org/dc/elements/1.1/
- http://ns.adobe.com/xap/1.0/
- http://ns.adobe.com/xap/1.0/mm/

## 9. Recommendations

- Treat with caution ? do not open if the source is untrusted.
- Scan with a second AV engine or upload to VirusTotal.
- Consider opening in an isolated sandbox environment.
- Inspect embedded URLs and JavaScript if safe to do so.

*DISCLAIMER: This report is generated by an automated static-analysis system. It is intended as a screening tool only and does not replace expert forensic analysis. False positives and false negatives are possible. Do not rely solely on this report for security decisions.*